

PASSAGE ISLAND, MI, USA

WMO: 994090

Lat: 48.223N

Lon: 88.366W

Elev: 641

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.3	(c) -4.8	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 44.0	(k) 15.9	(l) 41.0	(m) 17.0	(n) 19.4	(o) 330	(p) 0.905

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.4	(c) 69.1	(d) N/A	(e) 67.2	(f) N/A	(g) 65.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 15.4	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 41.1	(b) 35.5	(c) 32.1	(d) DB	(e) -12.7	(f) 73.2	(g) 3.7	(h) 3.7	(i) -15.3	(j) 75.9	(k) -17.5	(l) 78.0	(m) -19.6	(n) 80.1	(o) -22.3	(p) 82.8	(q) N/A
(4)			(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	14.0	16.8	24.8	34.7	42.3	48.8	56.3	60.8	56.0	45.0	33.3	24.1
	DBStd	17.00	11.45	10.75	9.29	5.57	4.83	4.17	5.69	4.30	5.42	6.42	8.60	9.63
	HDD50	5106	1116	929	780	460	246	72	9	0	13	180	500	801
	HDD65	9799	1581	1349	1245	910	704	487	273	142	273	619	950	1266
	CDD50	798	0	0	0	0	7	35	205	334	192	25	0	0
	CDD65	17	0	0	0	0	0	0	4	11	2	0	0	0
	CDH74	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	15.6	18.6	18.0	16.2	15.1	13.8	12.0	12.2	13.3	14.7	17.4	18.6	17.9
	PrecAvg	34.6	1.9	1.3	1.7	2.6	3.5	3.9	3.8	3.5	4.1	3.6	2.7	2.1
(15) Precipitation	PrecMax	41.3	4.4	3.3	3.3	7.7	8.1	8.0	6.7	5.4	8.5	6.6	5.2	3.7
	PrecMin	25.4	0.4	0.4	0.4	0.7	1.9	1.8	2.2	1.7	1.7	1.3	1.0	1.0
	PrecStd	4.0	0.7	0.7	0.6	1.3	1.2	1.4	1.1	1.0	1.5	1.3	1.0	0.7
	PrecStd	4.0	0.7	0.7	0.6	1.3	1.2	1.4	1.1	1.0	1.5	1.3	1.0	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	38.5	38.6	46.1	50.2	61.4	65.1	71.5	71.6	69.2	61.4	49.9	41.9	
	0.4% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2% DB	36.1	35.7	39.9	45.5	56.2	61.2	68.3	69.5	66.4	58.0	47.3	39.9	
	2% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% DB	33.9	33.3	37.1	42.5	52.0	58.2	66.2	67.9	64.2	55.5	45.9	38.2	
	5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10% DB	31.2	31.4	35.1	40.5	48.5	55.2	64.3	66.2	62.3	53.3	44.3	36.6	
	10% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	0.4% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% DB	11.9	10.8	8.6	7.2	9.1	12.0	10.5	6.4	6.2	6.0	7.5	9.1	
	5% DB MCDBR	11.7	10.8	8.9	10.3	15.9	15.9	11.5	8.0	8.1	7.7	6.9	8.4	
	5% DB MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% DB MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(35) Mean Daily Temperature Range	5% WB MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(40) Clear-Sky Solar Irradiance	taub	0.276	0.274	0.304	0.337	0.362	0.375	0.397	0.373	0.360	0.321	0.293	0.268	
	taud	2.464	2.461	2.414	2.392	2.422	2.413	2.407	2.475	2.485	2.554	2.539	2.518	
	Ebn at Noon	258	286	292	290	285	281	273	276	269	264	250	249	
	Edh at Noon	20	25	30	35	35	36	36	32	29	23	18	17	
(44) All-Sky Solar Radiation	RadAvg	332	648	1077	1502	1743	1882	1897	1632	1169	689	356	263	
	RadStd	39	69	98	155	116	129	104	121	78	85	39	27	

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page